

Bachelor of Science in Civil Engineering

Year of study: 2nd Year; 2nd Semester

Course Title: CIVIL ENGINEERING DRAWING AND
DESIGN

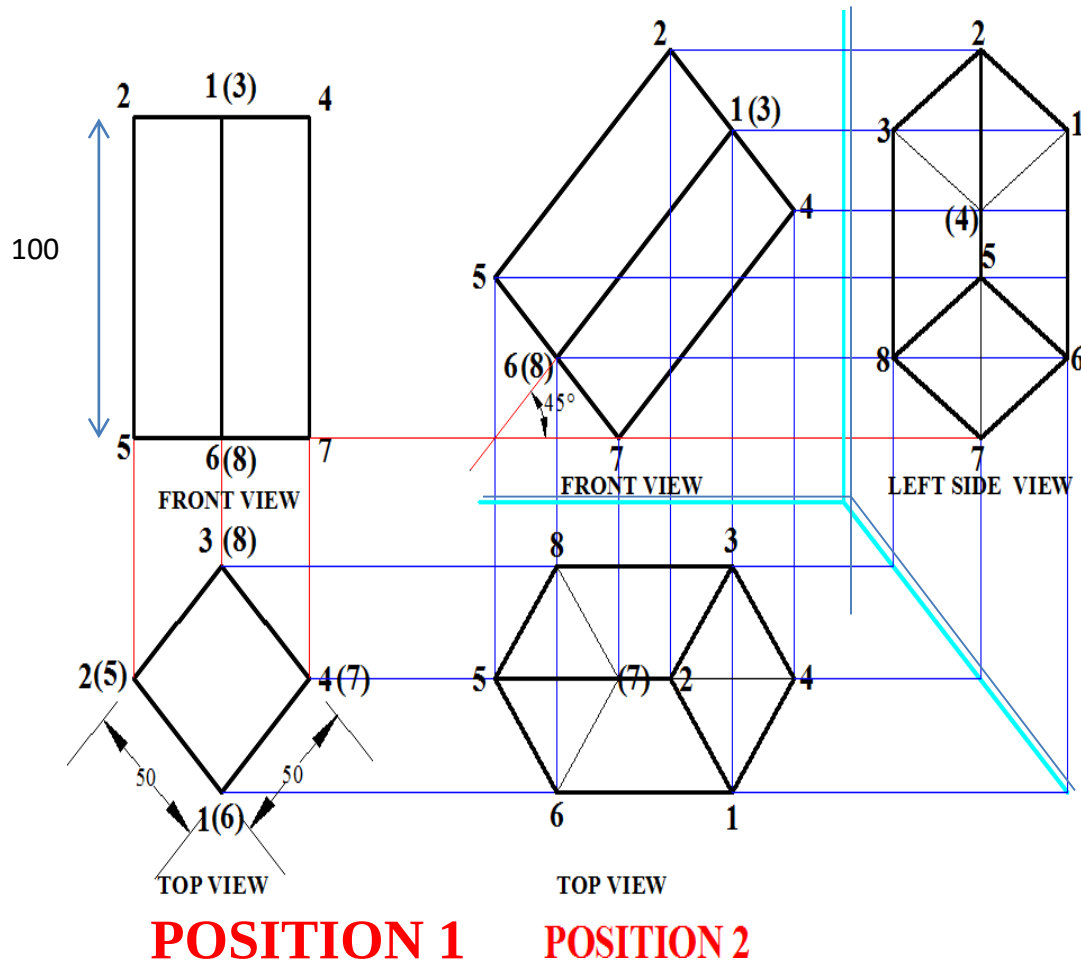
Course Code: FCC3 242

TUTORIAL ONE

PROJECTIONS OF SURFACES AND SOLIDS

(a) Square prism

A square prism is oriented as shown in Position 1. The prism is then tilted in the clockwise direction such that the plane 1-3-8-6 makes an angle of 45° to the horizontal as shown in Position 2. Draw the Front View, Left Side View and Top View in this position.

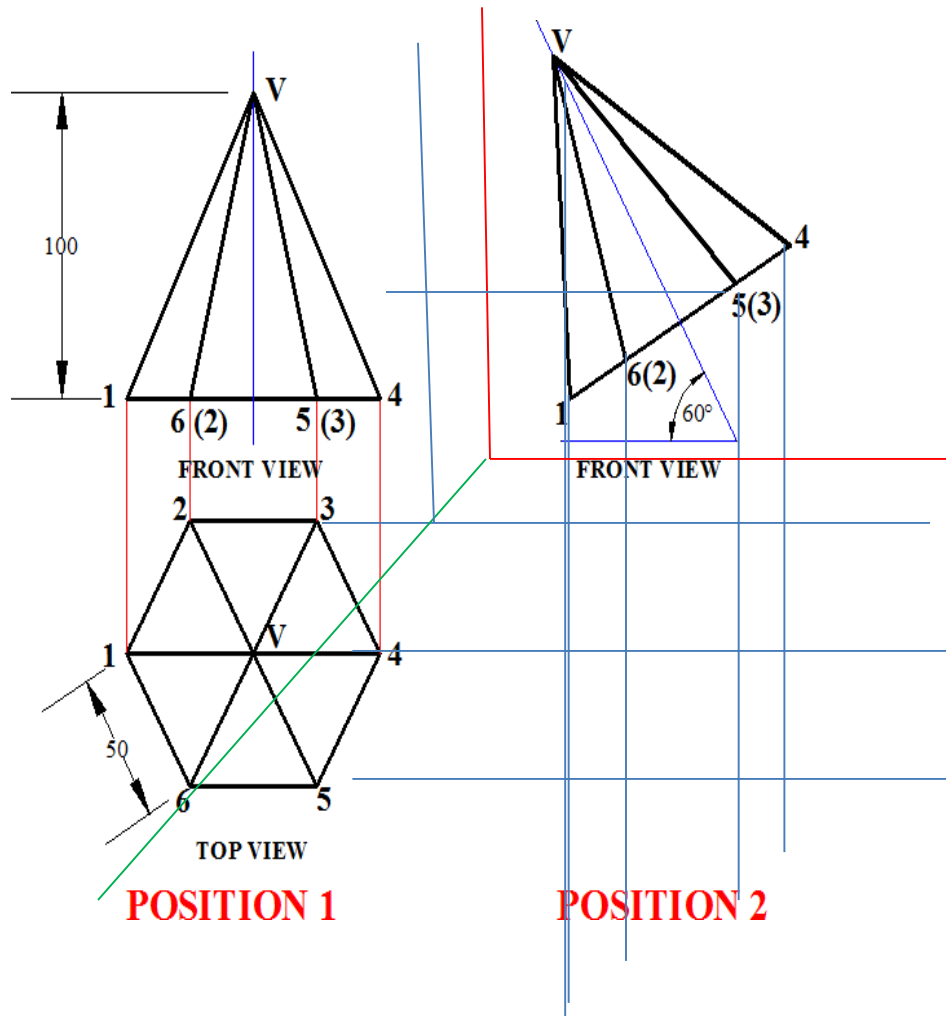


(b) Hexagonal right pyramid

The Front and Top Views of a (right) hexagonal pyramid are shown in Position 1 below. The pyramid is tilted in the anti-clockwise direction such that its vertical axis makes 60° to the horizontal as shown in Position 2.

(i) Draw the Front View and Top View in position 1.

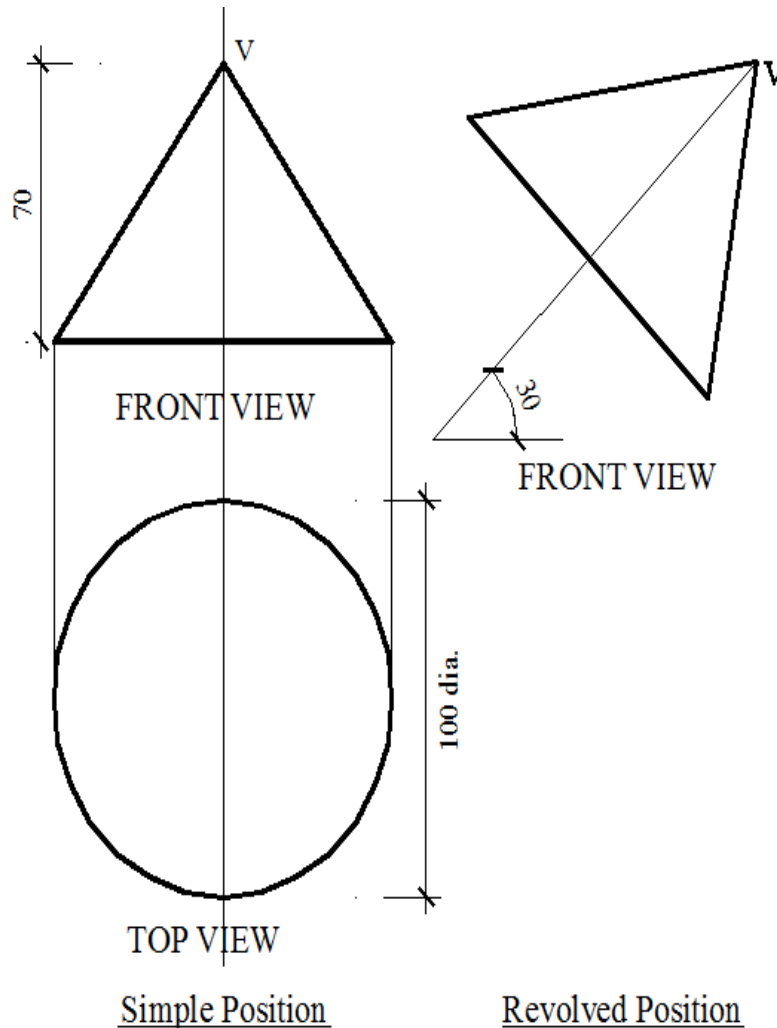
(i) Draw the Front View and Top View and Right Side View in position 2.



(c) Projections of a cone

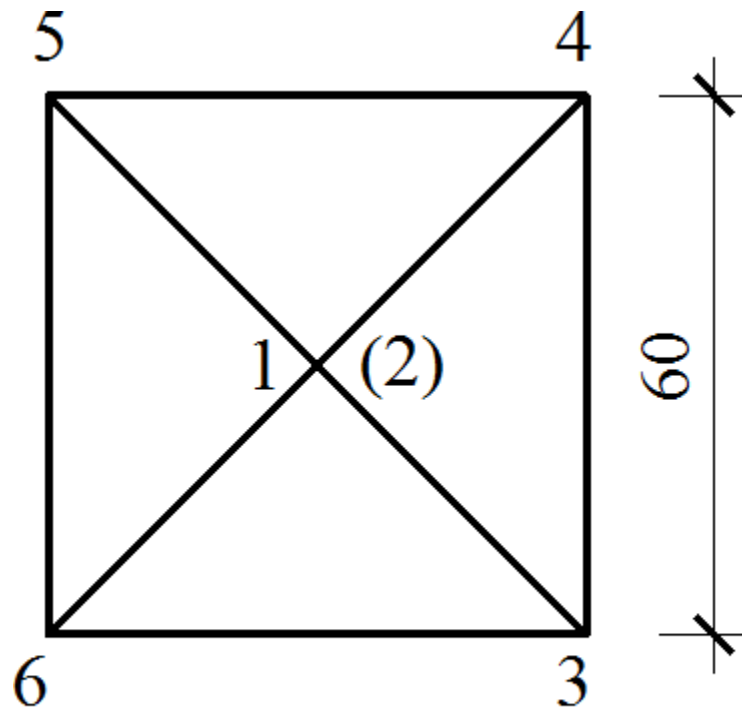
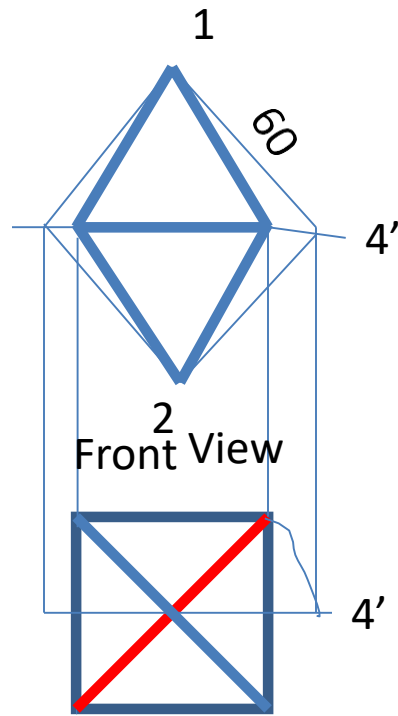
For the two orientations of the cone,

- (i) Draw the Front view and Top View of the simple position
- (ii) Draw the front View, Top View and Left Side View of the revolved position



(d) Projections of an octahedron

- (a) An octahedron is oriented in the simple position shown in the Top View below. In this position draw the Front View and the Top View.
- (b) The Front View in (a) of the octahedron is revolved in the clockwise direction such that at least one of the sloping surfaces is parallel to the Horizontal Picture Plane. Draw this front View and the Top View.

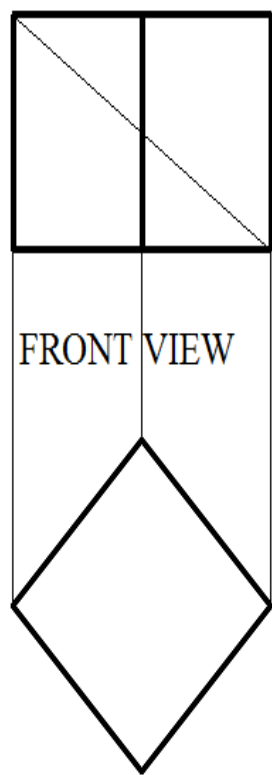


Top View

TOP VIEW

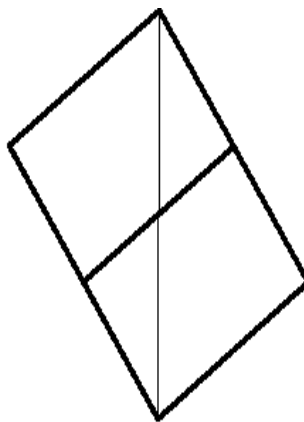
(e) Projections of a cube

- (a) A cube of side 50mm is oriented in the simple position as shown below. In this position draw Front View and Top View.
- (b) The Front View in (a) is revolved in such a way that one of the solid diagonals is vertical. In this position draw the Front View and the Top View.



TOP VIEW

Simple Position



FRONT VIEW

Revolved Position

**TO BE SUBMITTED ON 7TH
MARCH 2025 AT 2.00 PM.**

Drawing Instruments

Pencil e.g. HB, H series, pencil leads

A3 paper: 420 x 297 landscape

Masking tape

Scale ruler

T-square

Set squares 45 and 30-60 degrees

Protractor

Pair of compasses,

Dividers

Calculator

Eraser

AutoCAD